

4.2.3 For Statement

In this case, the condition of the loop is first checked. If it holds true, only then is the body of the loop executed. The syntax of this loop is as follows:

```
for(initialization;condition;increment or decre-
ment)
{
    statement 1;
    statement 2;
    ----- ;
    statement n;
}
```

Here, 'n' is a positive integer. Initialization refers to the initialization of the variable. This counts the number of times the body of the loop is executed. Condition refers to whether the body of the loop is executed or not. If this condition holds true, the body of the loop is executed. Increment or decrement is the part that increases or decreases the value of the variable discussed above. The variable should be declared earlier as other variables will be initialized by some value and then the condition will be checked. If the condition returns true, the body of the loop will be executed. Finally, before rechecking the condition, the value of the variable is increased or decreased as per requirement. The initialization takes place only once. After that, this portion will not be executed regardless the number of times the body of the loop execute.

The following program will display only the first 10 integers. We have done this before using 'while' loop, but this will help us to understand how a 'for' loop works.